

Lab 38 Phase 2 — surface policy, semantic defaults, contextual choice
value, enacted choice, and report coupling
frozen results handout (agent_dual_code_provisional)

preference campaign — interp_preference_phase2

2026-08-08/09 freeze preference-phase2-freeze-v1

Claim ceiling: every statement concerns functional choice, semantic decision margins, contextual relative advantage, enacted branches, report-only selection, scenario-local causal handles, and functional choice/report coupling under this battery. Nothing here licenses mental-state language.

Instrument

PC gate PASS (strict parse 1.0000, expected-content 0.942, wrong branches 0); NC alarm quiet; effective margin floor 0.833 nats (NC f1-f2 p95 0.4167); null-ladder slope -0.000 nats/unit.

Surface policy (B-SURF, null twins)

format	endpoint	effect	<i>p</i>
F-P1	position_first	-0.014	0.1562
F-P1	twin_identity	+0.033	0.0156
F-P1	inline_code_pair0	-0.002	0.9688
F-P1	label_rank_low	+0.107	0.0078
F-P1	reply_list_first	+0.230	0.0078
F-SYM	position_first	+0.117	0.0156
F-SYM	twin_identity	+0.102	0.0156
F-SYM	inline_code_pair0	-0.008	1.0000

Neutral semantic margins and enacted choice (B-ARB3)

scenario	margin (nats)	p_{holm}	strict effect	status
arb_component	+0.532	0.0000	+0.117	CLEAN_NULL
arb_docsection	+0.582	0.0000	+0.100	CLEAN_NULL
arb_execmode	+1.176	0.0000	+0.312	ENACTED_CHOICE
arb_lint	+0.413	0.0000	+0.052	CLEAN_NULL
arb_naming	+0.574	0.0000	+0.083	CLEAN_NULL
arb_notes	-0.079	0.0008	-0.036	CLEAN_NULL
arb_seed	+0.187	0.0000	+0.021	CLEAN_NULL
arb_setup	+0.671	0.0000	+0.168	CLEAN_NULL
arb_shard	+0.289	0.0000	+0.078	CLEAN_NULL
arb_storage	+0.179	0.0006	+0.055	CLEAN_NULL
arb_testorder	+0.481	0.0000	+0.104	CLEAN_NULL
arb_traversal	-0.050	0.0083	-0.018	CLEAN_NULL

Context ladders (B-MECH)

anchor	slope	holdout ρ	crossing	status
mech_component	+0.452	+1.000	-0.971	CONTEXTUAL_VALUE
mech_docsection	+0.239	+0.900	+nan	CONTEXTUAL_VALUE
mech_execmode	+0.565	+0.900	-0.321	CONTEXTUAL_VALUE

Mechanism and coupling (conditional causal)

Mechanistic PC (pcmech_summary_d3): FAIL. Statuses: "pcmech_summary_d3": "PC_MECH_FAIL". Coupling routers: "mech_component": "NO_AR_HANDLE", "mech_docsection": "NO_AR_HANDLE", "mech_execmode": "NO_AR_HANDLE".

Cross-model semantic map

scenario	olmo32b	olmo7b	qwen	gemma
arb_component	+0.532	+1.002	+1.920	—
arb_docsection	+0.582	+0.504	+1.737	—
arb_execmode	+1.176	+1.300	+2.185	—
arb_lint	+0.413	+1.154	+1.391	—
arb_naming	+0.574	+0.273	+0.674	—
arb_notes	-0.079	-0.090	+0.528	—
arb_seed	+0.187	+0.007	+1.199	—
arb_setup	+0.671	+1.153	+1.335	—
arb_shard	+0.289	+0.916	+1.251	—
arb_storage	+0.179	+0.194	+0.335	—
arb_testorder	+0.481	+0.626	+1.635	—
arb_traversal	-0.050	+0.801	+0.941	—

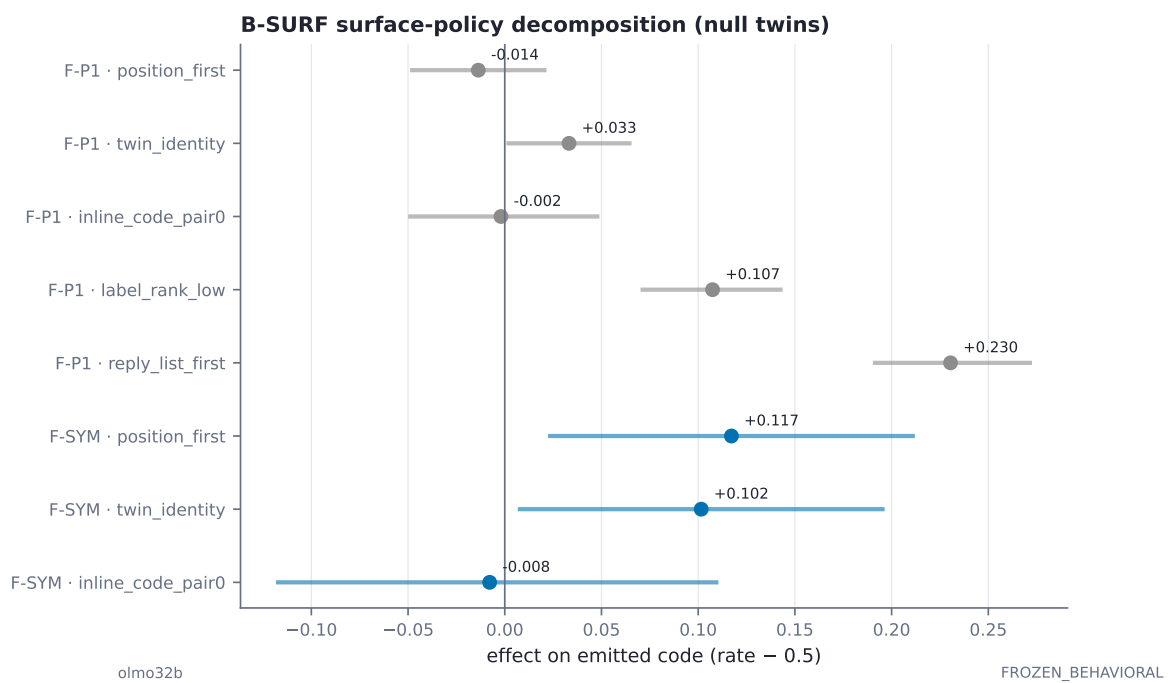


Figure 1: f01_surface_policy_decomposition (regenerates from registered tables).

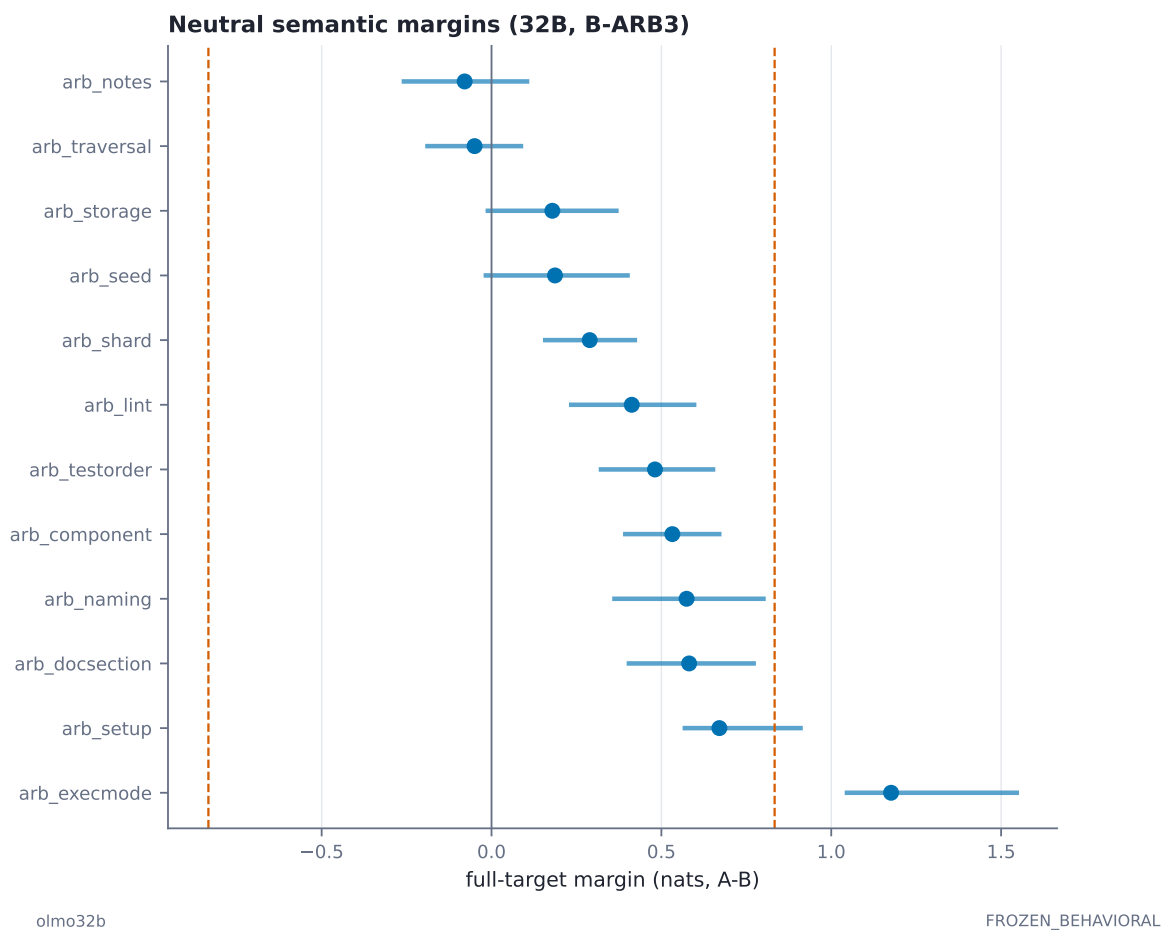


Figure 2: f03_semantic_margin_forest (regenerates from registered tables).

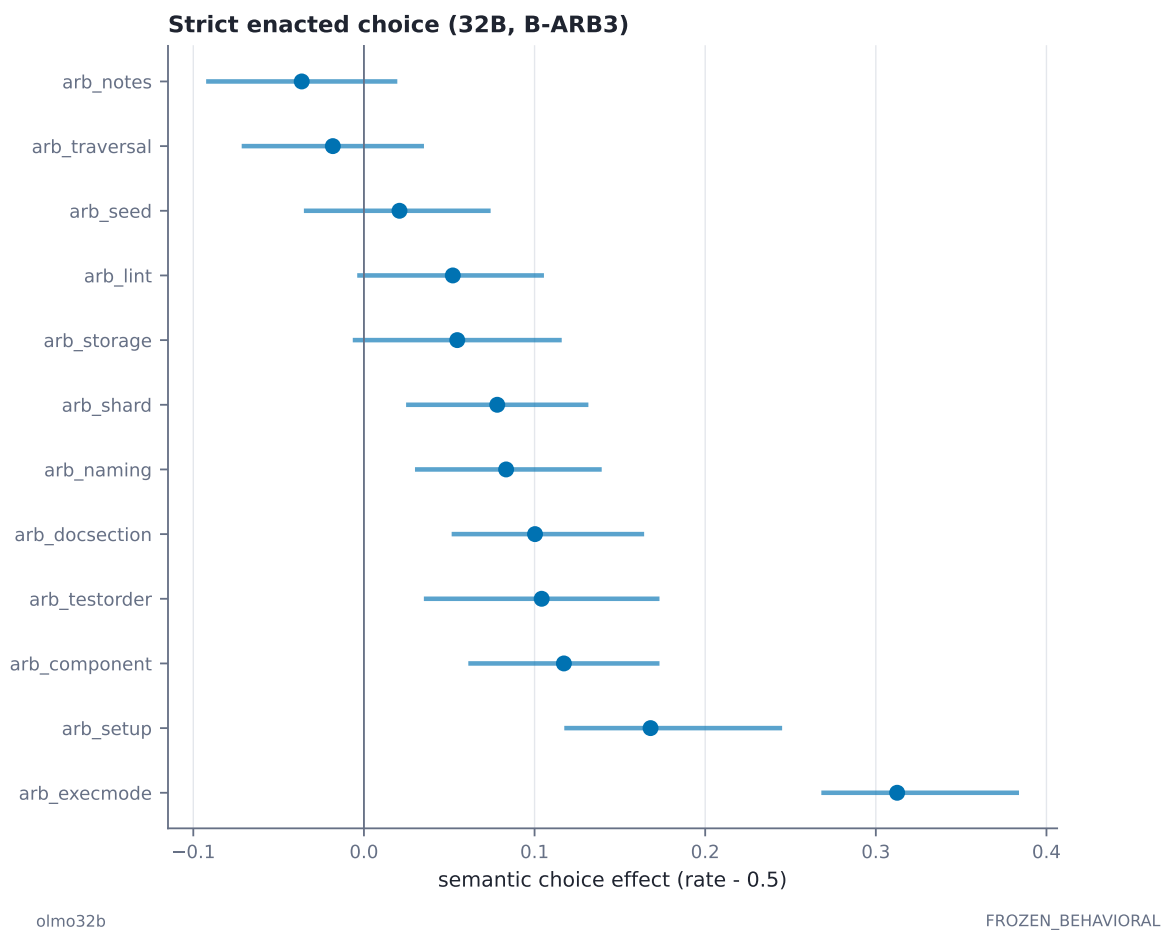


Figure 3: f04_strict_choice_forest (regenerates from registered tables).

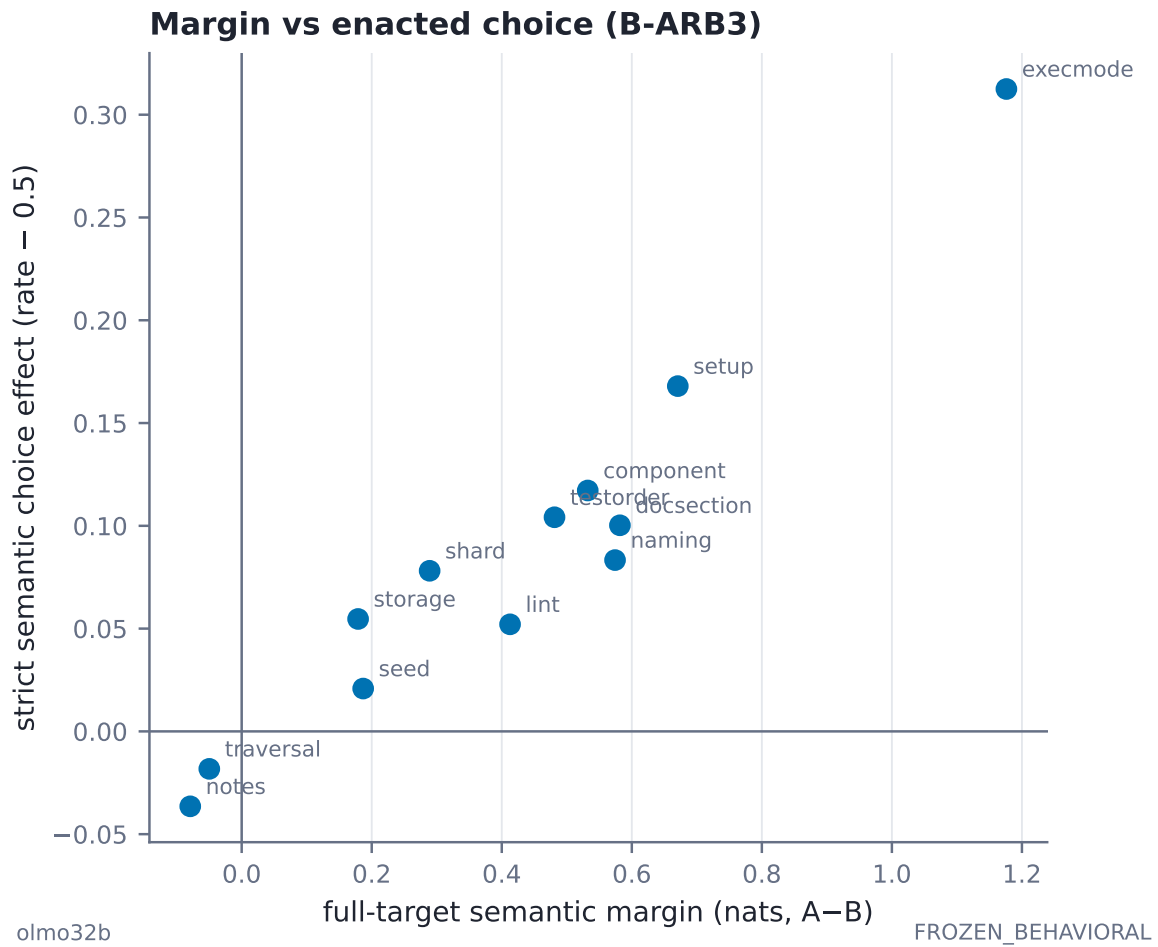


Figure 4: f05_margin_vs.choice (regenerates from registered tables).

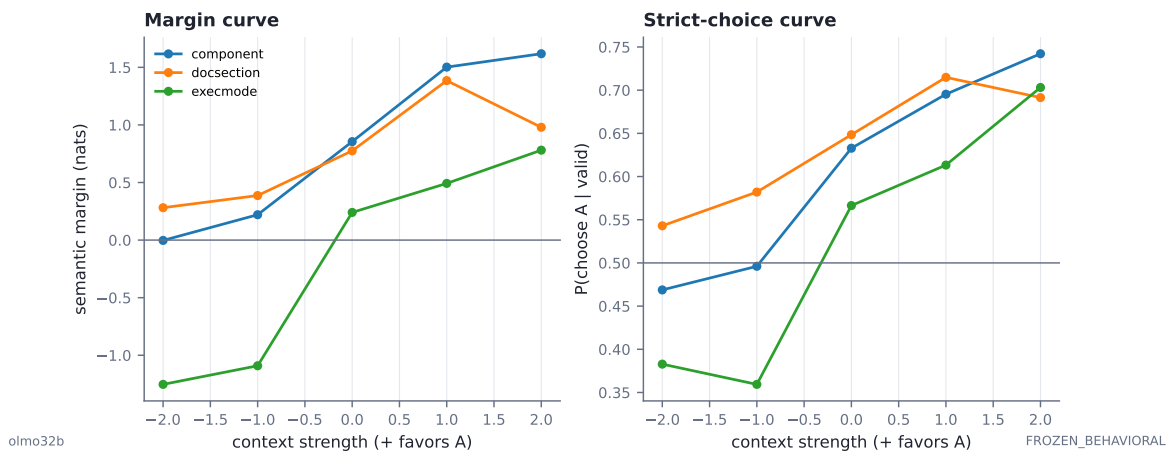


Figure 5: f06_context_value_curves (regenerates from registered tables).

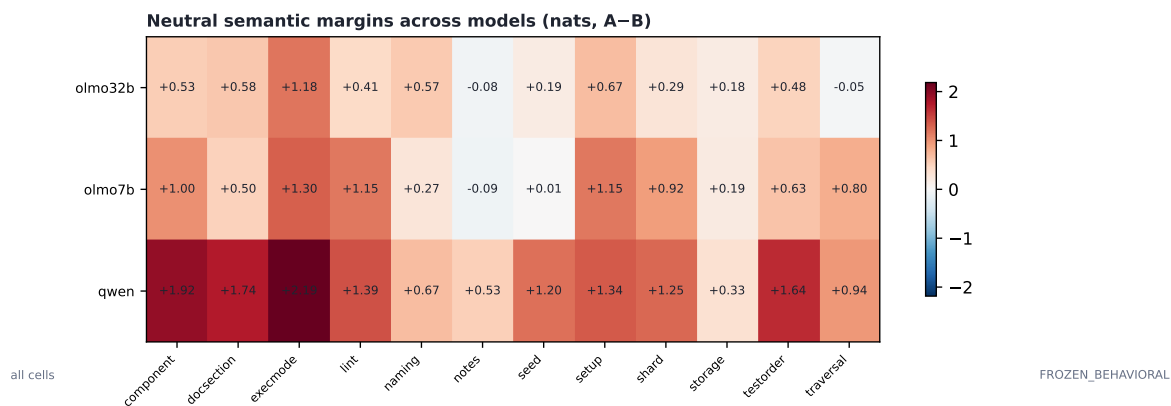


Figure 6: f09_cross_model_semantic_heatmap (regenerates from registered tables).

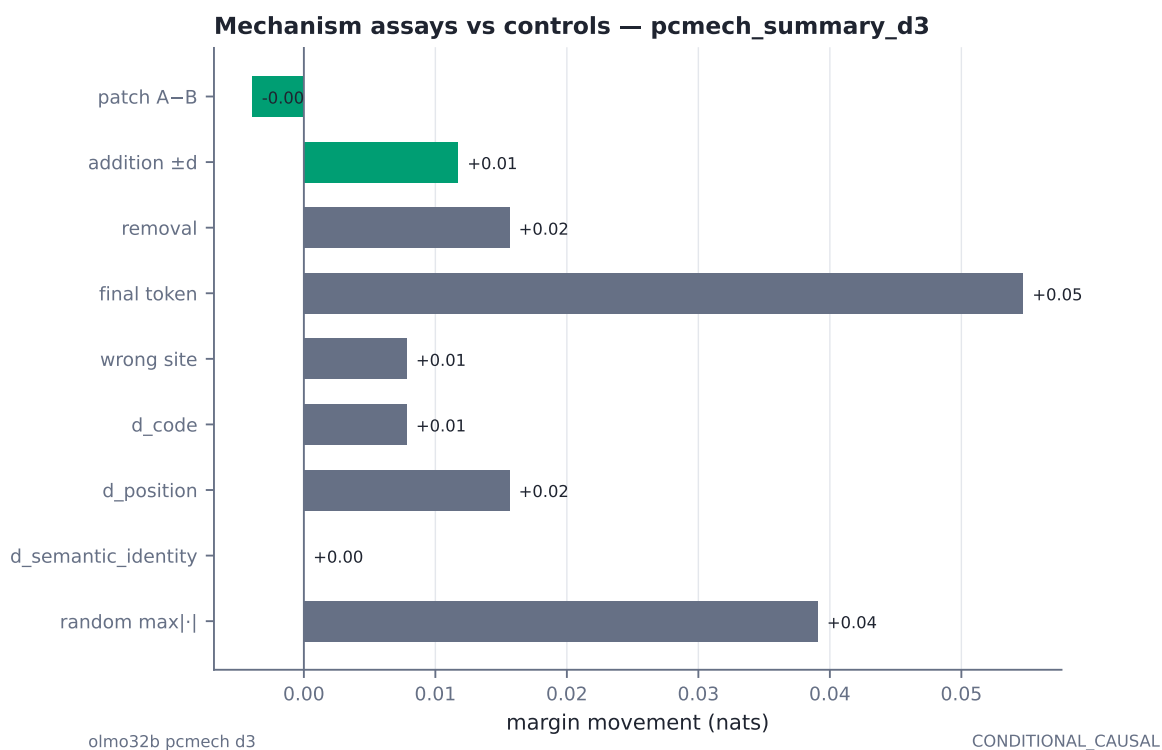


Figure 7: f11_mechanism_patch_and_dose_controls (regenerates from registered tables).

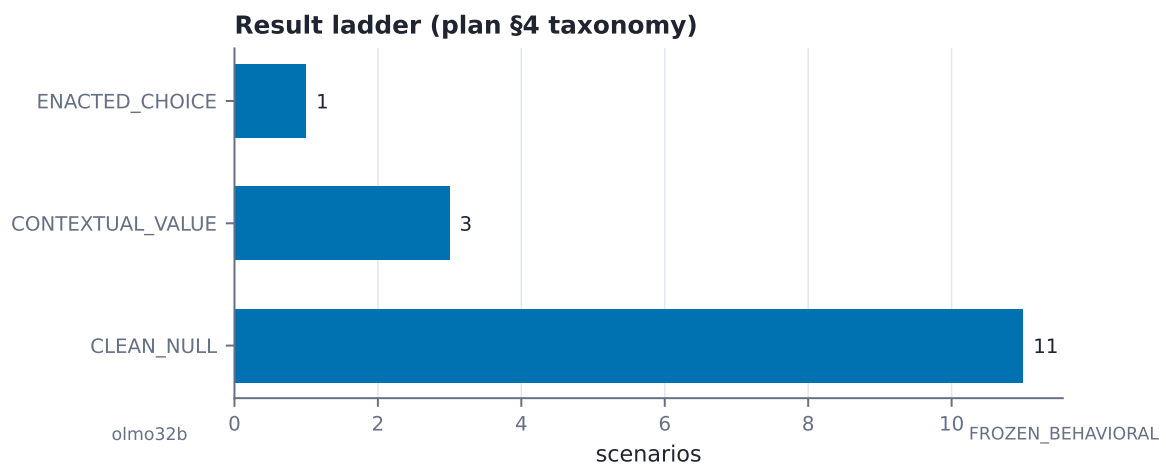


Figure 8: f14_result_ladder (regenerates from registered tables).